Networks in their Surrounding Contexts

EM1421 - Social media and web analytics

{Slides courtesy of Prof. Fabiana Zollo}

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Introduction

- Each individual in a social network has a distinctive set of personal characteristics
- Similarities between two people can strongly influence link formation
- **Surrounding contexts**: factors that exist outside the nodes and edges of a network, but affect how the network evolves

Homophily

- **Homophily**: the principle that we tend to be similar to our friends
- Typically, your friends don't look like a random sample of the underlying population
- Viewed together, they're generally similar to you
  - in age;
  - along racial and ethnic dimensions;
  - in characteristics such as where they live, work, their occupation, their interests, their beliefs

Homophily

- Most of us have friends that cross all these boundaries
- But in aggregate **links in a social network tend to connect people who are similar to one another**
- Homophily can drive the formation of link in a social network

Homophily

- We motivated triadic closure by hypothesizing intrinsic mechanisms (trust, opportunity, incentive)
- Social contexts also provide a natural bases for triadic closure:
  - Since A-B and A-C are friends, homophily suggests that B and C are similar to each other
  - There is an elevated chance the B-C friendship will form, even if they're not aware of having A as a common friend

Measuring Homophily

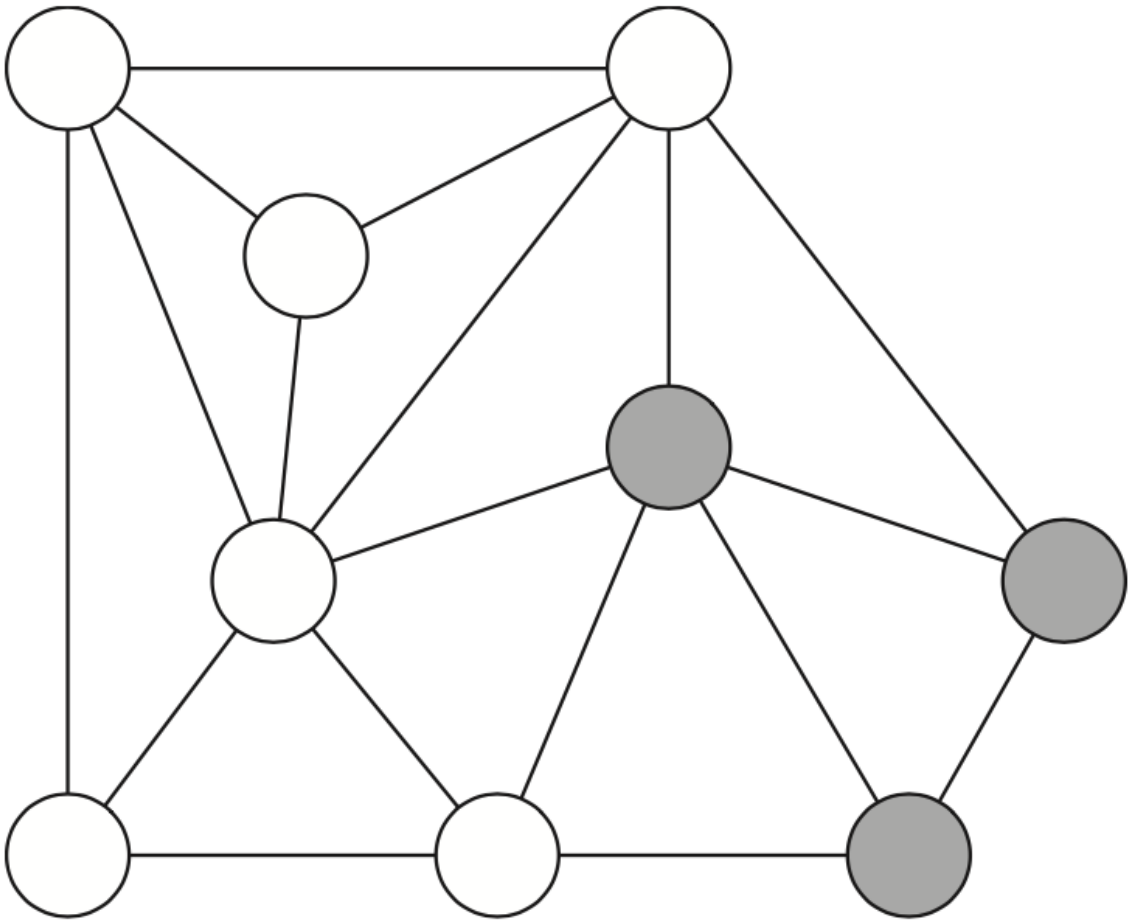
Given a particular characteristic of interest (like age), is there a simple test that can be applied to estimate whether a network exhibits homophily according to this characteristic?

Measuring Homophily

- Suppose that we have the friendship network of a school classroom
- Hypothesis: the network exhibits homophily by gender
  - Boys tend to be friends with boys
  - Girls tend to be friends with girls

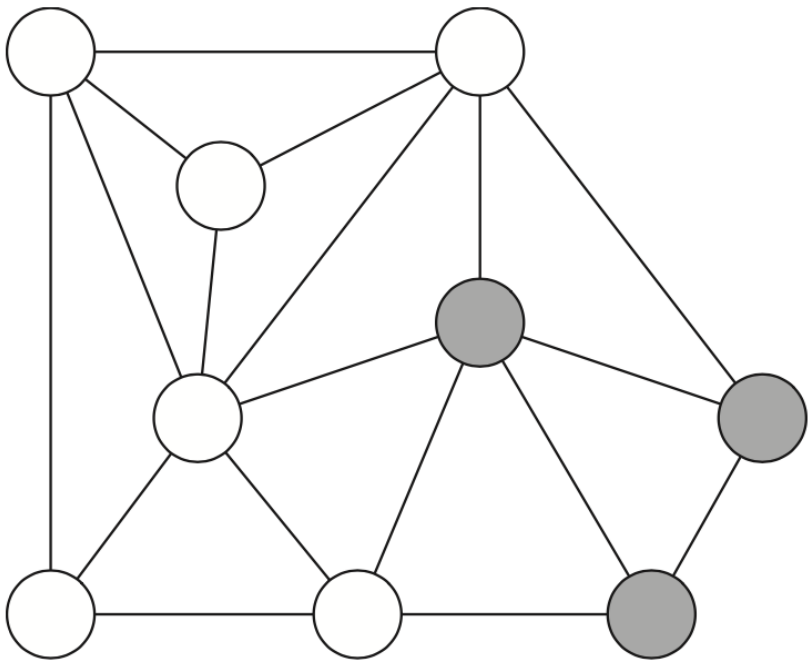
Measuring Homophily

- Shaded nodes = girls
- White nodes = boys



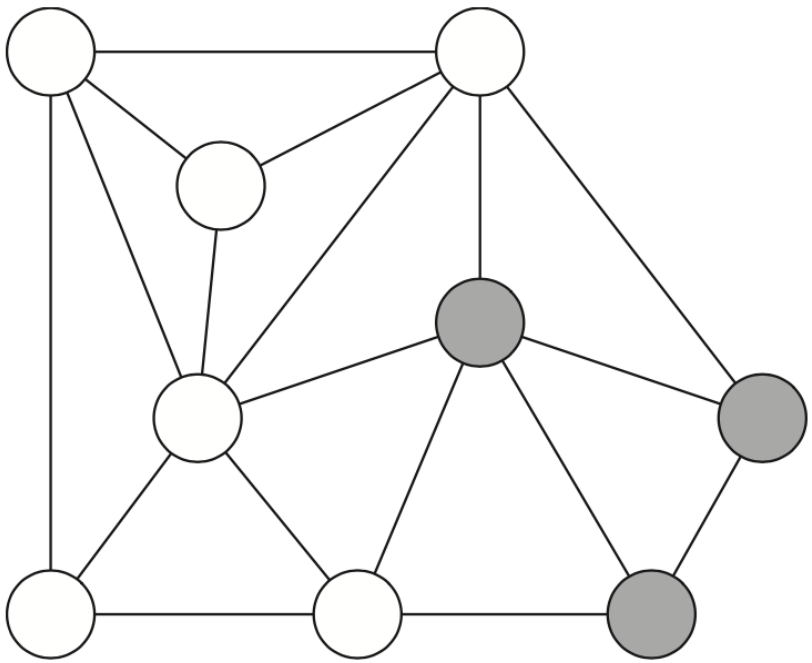
Measuring Homophily

- If there were no cross-gender edges at all, the answer would be straightforward
- But in real networks we expect a more subtle effect that is visibile mainly in aggregate



Measuring Homophily

- Is this network consistent with homophily?
- Any idea?



Measuring Homophily

- There is a natural numerical measure that we can use to address this question
- What would it mean for a network **not** to exhibit homophily by gender?

Measuring Homophily

- It would mean that the proportion of male and female friends that a person has looks like the background male/female distribution in the full population
  Definition of "no-homophily": If we were to randomly assign each node a gender according to the gender balance in the real network, then the number of the cross-gender edges should not change significantly relative to what is seen in the real network
- With no homophily, friendships are being formed as though random mixing were occurring

Measuring Homophily

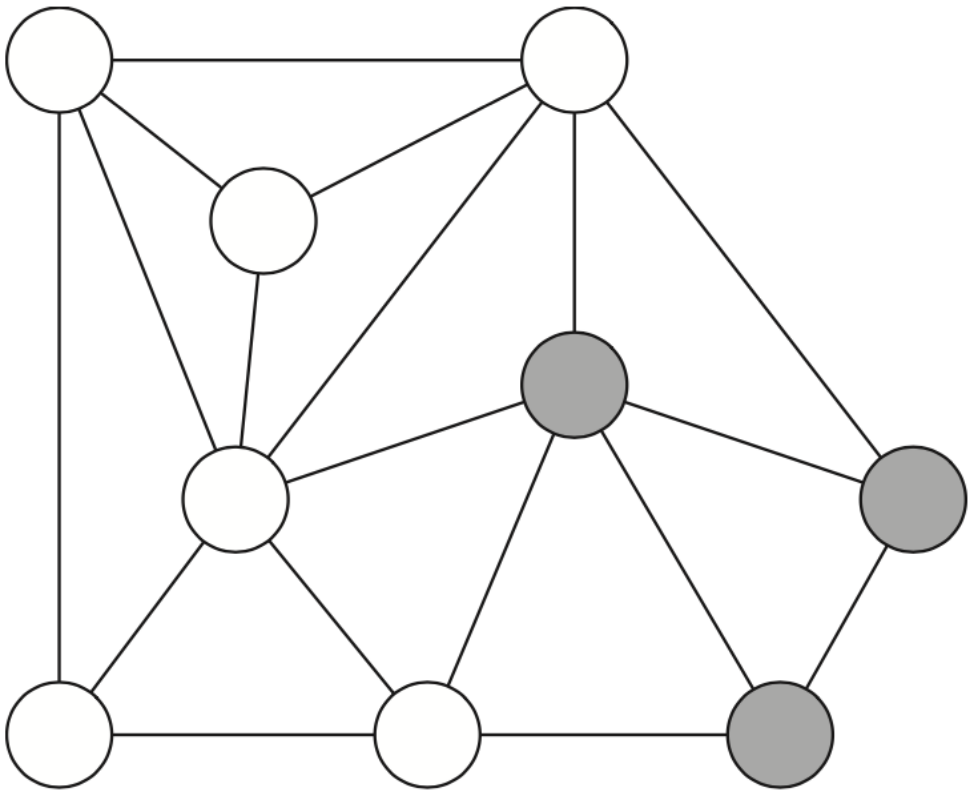
- Suppose we have a network in which
  - A fraction p of all individuals are male
  - A fraction q of all individuals are female
- Consider a given edge in this network
- If we assign each node the gender *male* with probability p, and the gender *female* with probability q, then both ends of the edge will be male with probability p^2 and both ends female with probability q^2
- A cross-gender edge happens with probability $2pq$

Measuring Homophily

- Our test can be summarized as follows:
  Homophily Test: If the fraction of cross-gender edges is significantly less than $2pq$, then there is evidence for homophily.

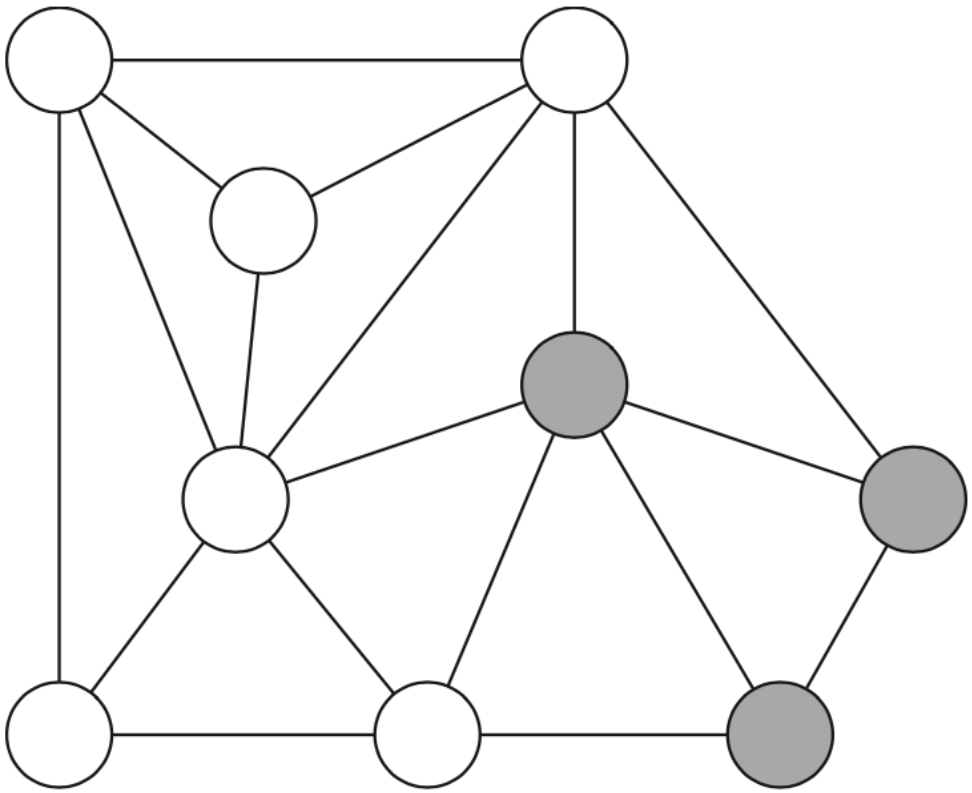
Measuring Homophily

- $p = \frac{6}{9} = \frac{2}{3}$
- $q = \frac{3}{9} = \frac{1}{3}$



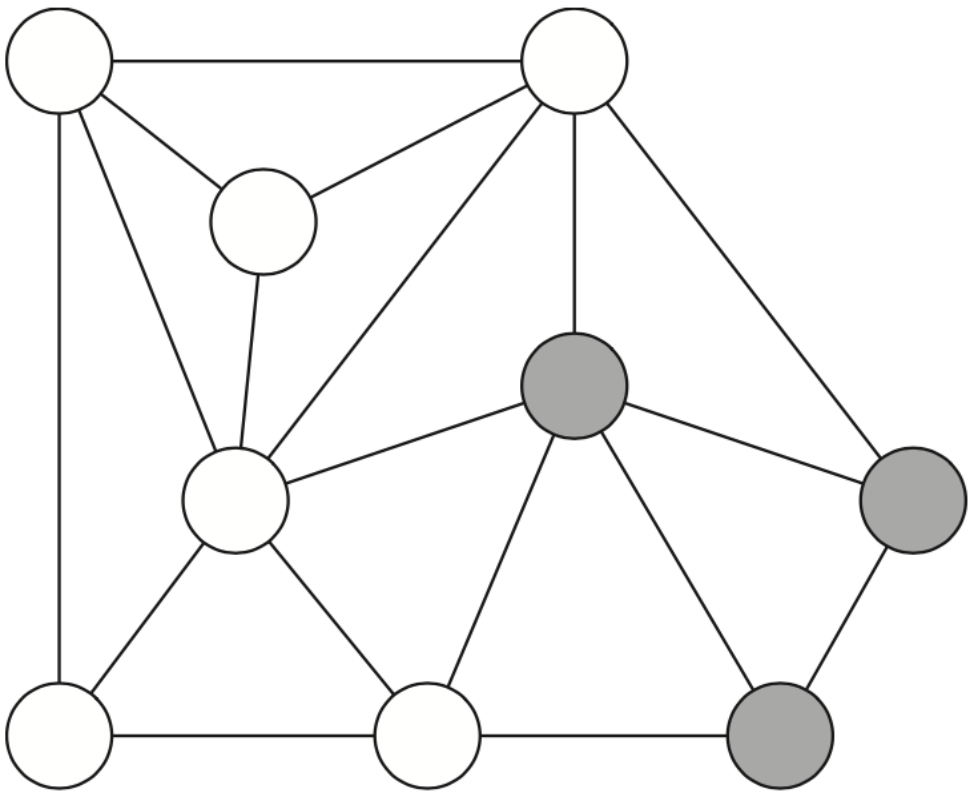
Measuring Homophily

- $p = \frac{6}{9} = \frac{2}{3}$
- $q = \frac{3}{9} = \frac{1}{3}$
- $\frac{5}{18}$ edges are cross-gender



Measuring Homophily

- $p = \frac{6}{9} = \frac{2}{3}$
- $q = \frac{3}{9} = \frac{1}{3}$
- $\frac{5}{18}$ edges are cross-gender
- $2pq = 2 * \frac{2}{3} * \frac{1}{3} = \frac{4}{9} = \frac{8}{18}$



Measuring Homophily

- In other words, with no homophily we should expect to see 8 cross-gender edges instead of 5
- We thus found some evidence of homophily

Measuring Homophily

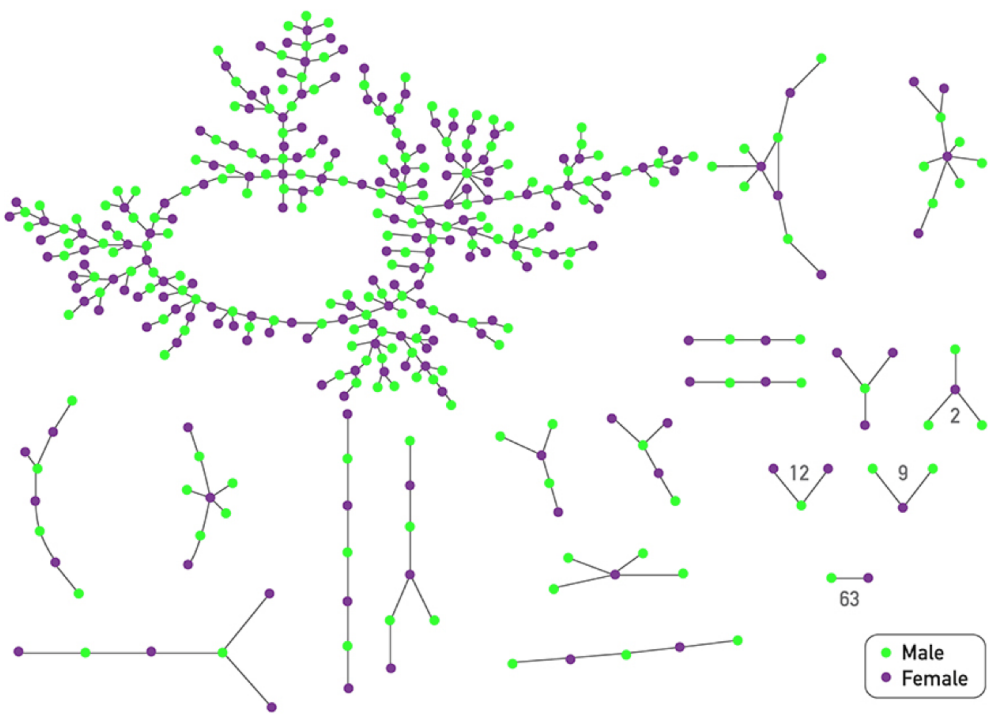
Some notes (1/3)

- The number of cross-gender edges in a random assignment can deviate from the expected value of $2pq$
- To perform the test in practice, one needs a definition of "significantly less of" (e.g., quantifying the significance of a deviation below the mean)

Measuring Homophily

Some notes (2/3)

- It is possible for a network to have a fraction of cross-gender edges that is **significantly more than** $2pq$
- We say that the network exhibits **inverse homophily**



Measuring Homophily

Some notes (3/3)

- It is easy to extend the homophily test to other characteristics
- When the characteristic can only take two possible values (e.g. political orientation, native language) it's straightforward
- What about when the characteristic can take on more than two possible values?



Measuring Homophily

Some notes (3/3)

- What about when the characteristic can take on more than two possible values?
- We say that an edge is **heterogeneous** if it connects two different nodes based on that characteristic
- We then ask the number of heterogeneous edges compared to what we'd see if we were to randomly assign values for the characteristic to all nodes in the network
  - How? Using the proportions from the real data as probabilities

Selection

*The term **selection** refers to the tendency of people to form friendships with others who are like, i.e., selecting friends with similar characteristics*

- Selection may operate at several different scales and with different levels of intentionality
- Sometimes people choose friends who are most similar from among a clearly delineated pool of potential contacts
- In other cases the social environment is already favoring opportunities to form friendships with others like oneself

Selection

- Immutable characteristics: a person's attributes are determined at birth and they play a role in how this person's connections are formed
- Mutable characteristics (e.g., behaviors, activities, interests, beliefs, opinions): the effect on the social network becomes significantly more complex
- The process of selection still operates, with individual characteristics affecting the connections that are formed
- But another process comes into play as well..



Social Influence

- People may modify their behaviors to bring them more closely into alignment with the behaviors of their friends
- **Social influence**: the existing social connections in a network are influencing the individual characteristics of the nodes
- Social influence can be viewed as the reverse of selection
  - Selection: the individual characteristics drive the formation of links
  - Social influence: the existing links in the network serve to shape people's (mutable) characteristics

The Interplay of Selection and Social Influence

- When we look at a single snapshot of a network, it can be very hard to sort out the distinct effects and contributions of selection and social influence
- *Have the people in the network adapted their behaviors to become more like their friends, or have they sought out people who were already like them?*
- Such questions can be addressed using longitudinal studies of a social network and analyze the behavioral changes that occur after changes in an individual's network connections, as opposed to the changes to the network that occur after an individual changes his or her behavior

The Interplay of Selection and Social Influence

- Example: Study of the processes that lead pairs of adolescent friends to have similar outcomes in terms of scholastic achievement and delinquent behavior such as drug use
- Empirical evidence confirms that teenage friends are similar to each other in their behaviors, for both selection and social influence effects
  - Teenagers seek out social circles composed of people like them
  - Peer pressure causes them to conform to behavioral patterns within their social circles
- But how do these two effects interact? Is one of them more strongly at work than the other?

The Interplay of Selection and Social Influence

- Cohen and Kandel suggested that while both effects are present in the data, social influence is actually more moderate
- Understanding these mechanisms can be crucial for identifying possible interventions
- For example, if illicit drug use displays homophily across a social network (with students showing a greater likelihood to use drugs when their friends do) we can implement a program that targets certain high-school students and influences them to stop using drugs
- If the observed homophily is based on some amount of social influence, such a program could have a broad impact across the social network, by causing the friends of these targeted students to stop using drugs as well

The Interplay of Selection and Social Influence

- But we must be careful!
- If the observed homophily is arising instead almost entirely from selection effects, then the program may not reduce drug use beyond the students it directly targets
- Why?



The Interplay of Selection and Social Influence

- As these students stop using drugs, they change their social circles and form new friendships with students who don't use drugs
- But the drug-using behavior of other students is not strongly affected
- This example shows how careful analysis is needed to distinguish among different factors contributing to an aggregate conclusion
- Even when people tend to be similar to their neighbors in a social network, it may not be clear why

The Interplay of Selection and Social Influence

Deeper research questions are needed:

- Why is the homophily present?
- How will its underlying mechanisms affect the further evolution of the network?
- How do these mechanisms interact with possible outside attempts to influence the behavior of people in the network?

Affiliation

- We can work with networks that contains both people and contexts as nodes
- This might help us get additional insight into some broad aspects of homophily
- We'll focus on how to represent the set of activities a person takes part in, and how these affect the formation of links
- *Activity* here is a very general notion, like
  - Being part of a particular company, organization, or neigborhood
  - Frequenting a particular place
  - Pursuing a particular hobby or interest

Affiliation

- Such activities, when shared between two people, tend to increase the likelihood that they will interact and hence form a link in the social network
- We refer to these activities as **foci**, i.e., focal points of social interaction Scott Feld, 1981

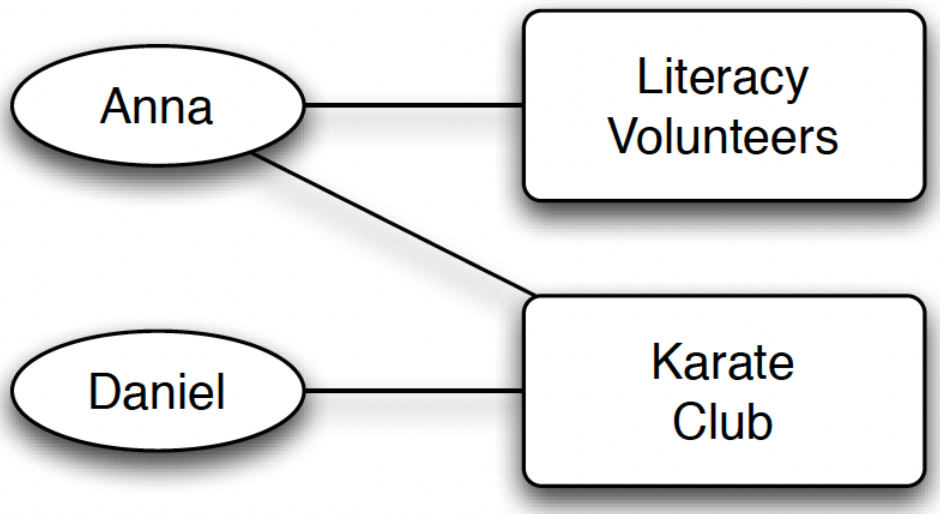
Affiliation Networks

- How can we represent the participation of a set of people in a set of foci using a graph?



Affiliation Networks

- We can use a bipartite graph, where we have a node for each person and a node for each focus
- We connect person A to focus X by an edge if A participates in X

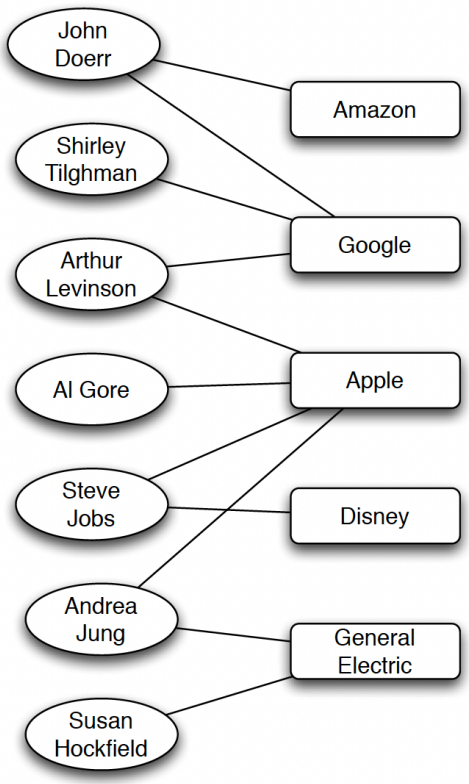


Affiliation Networks

- Affiliation networks are studied to understand the patterns of participation in structured activities
- Example: studying the composition of boards of directors of major corporations
- Boards of directors are relatively small advisory groups populated by high-status individuals
- Many people serve on multiple boards, and the overlaps in their participation can be naturally represented by an affiliation network

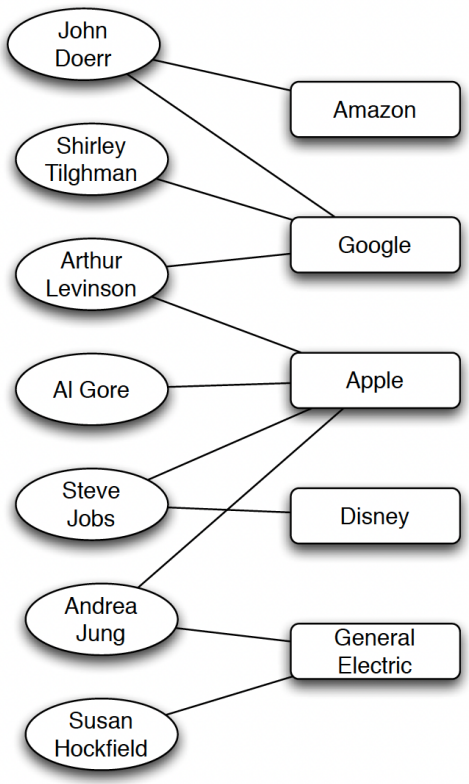
Affiliation Networks

- This affiliation network has the potential to reveal interesting relationships on both sides of the graph
- Two companies are implicitly linked by having the same person sit on both their boards (information and influence flows between companies)



Affiliation Networks

- Two people are implicitly linked by serving together on a board (patterns of social interaction among some of the most powerful members of society)
- Some context is still missing (the presidents of two major universities and a former Vice-President of the United States)



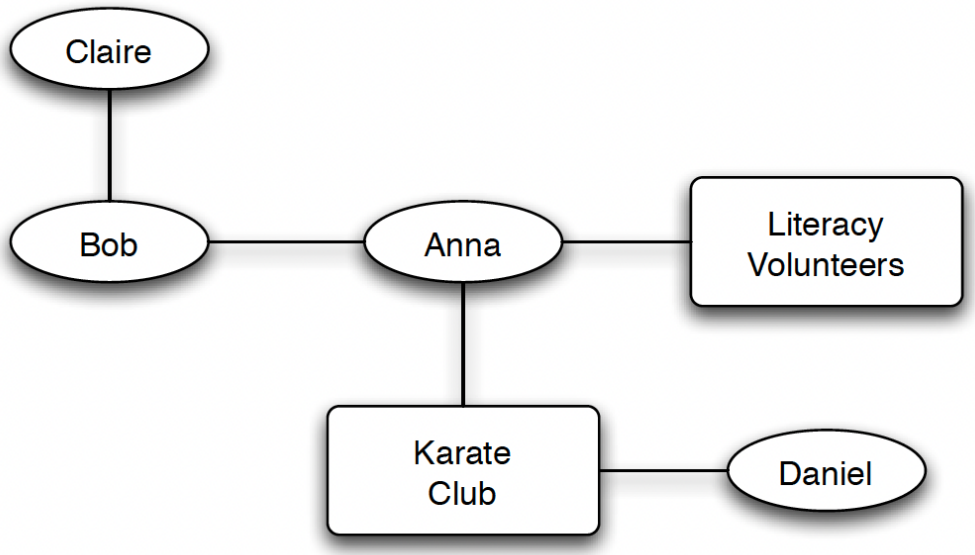
Co-Evolution of Social and Affiliation Networks

- Both social networks and affiliation networks change over time
- Co-evolution that reflects the interplay between selection and social influence:
  - If two people participate in a shared focus, this provides them with an opportunity to become friends
  - If two people are friends, they can influence each other's choice of foci

Co-Evolution of Social and Affiliation Networks

- We can use a network representation that extends the notion of an affiliation network
- We now introduce two distinct kinds of edges:
  - The first kind of edge connects two people and indicates a social relation
  - The second kind of edge connects a person to a focus, and indicates the participation of the person in the focus
- We call such a network a **social-affiliation network**

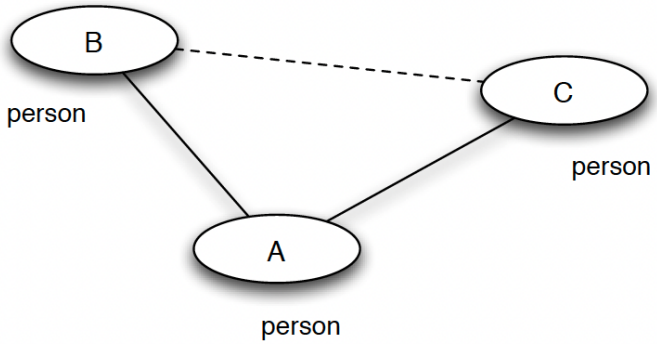
Co-Evolution of Social and Affiliation Networks



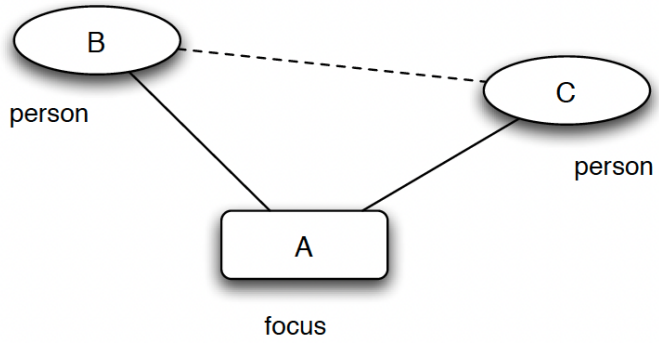
Co-Evolution of Social and Affiliation Networks

- We can study a range of different mechanisms for link formation, which can all be viewed as types of closure processes
- Suppose we have two nodes B and C with a common neighbor A in the network, and suppose that an edge forms between B and C
- There are several interpretations:
  - If A, B, and C each represent a person, then the formation of the link between B and C is **triadic closure**
  - If B and C represent people, but A represents a focus, this is the tendency of two people to form a link when they have a focus in common (**focal closure**)
  - If A and B are people, and C is a focus, then we have the formation of a new affiliation: B takes part in a focus that her friend A is already involved in (**membership closure**, kind of social influence)

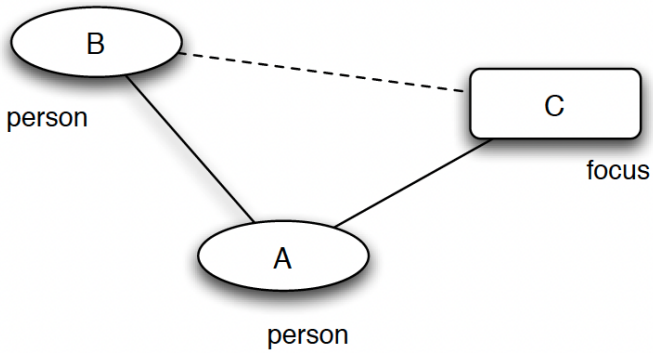
Co-Evolution of Social and Affiliation Networks



(a) *Triadic closure*



(b) *Focal closure*



(c) *Membership closure*

Tracking Link Formation: Triadic Closure

- How much more likely is a link to form between two people in a social network if they already have a friend in common *(i.e. triadic closure)*?
- How much more likely is an edge to form between two people if they have multiple friends in common?
- We can address these questions empirically using network data. Any idea?



Tracking Link Formation: Triadic Closure

1. We take two snapshots of the network at different times
2. For each k, we identify all pairs of nodes who have exactly k friends in common in the first snapshot, but who are not directly connected by an edge
3. We define $T(k)$ to be the fraction of these pairs that have formed an edge by the time of the second snapshot (empirical estimate for the probability that a link will form between two people with k friends in common)
4. We plot $T(k)$ as a function of k to illustrate the effect of common friends on the formation of links

Tracking Link Formation: Triadic Closure

- $T(0)$ is the rate at which link formation happens when it does not close a triangle
- $T(k)$ for larger k determine the rate at which link formation happens when it does close a triangle

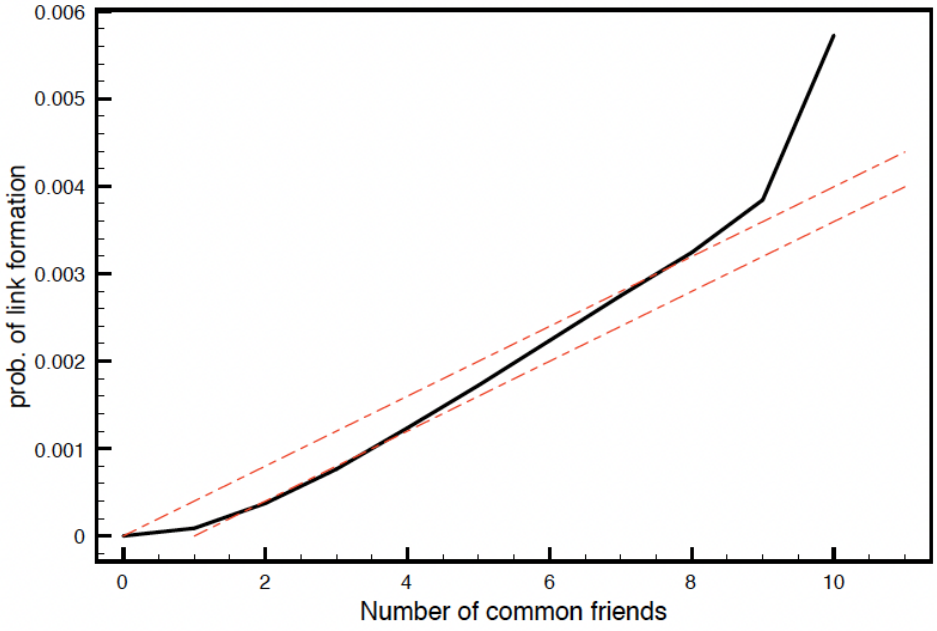
Tracking Link Formation: Triadic Closure

- Kossinets and Watts computed this function $T(k)$ using a dataset encoding the full history of e-mail communication among roughly 22,000 undergraduate and graduate students over a one-year period at a large U.S. university
- Kossinets and Watts built a network that evolved over time, joining two people by a link at a given instant if they had exchanged e-mail in each direction at some point in the past 60 days

Tracking Link Formation: Triadic Closure

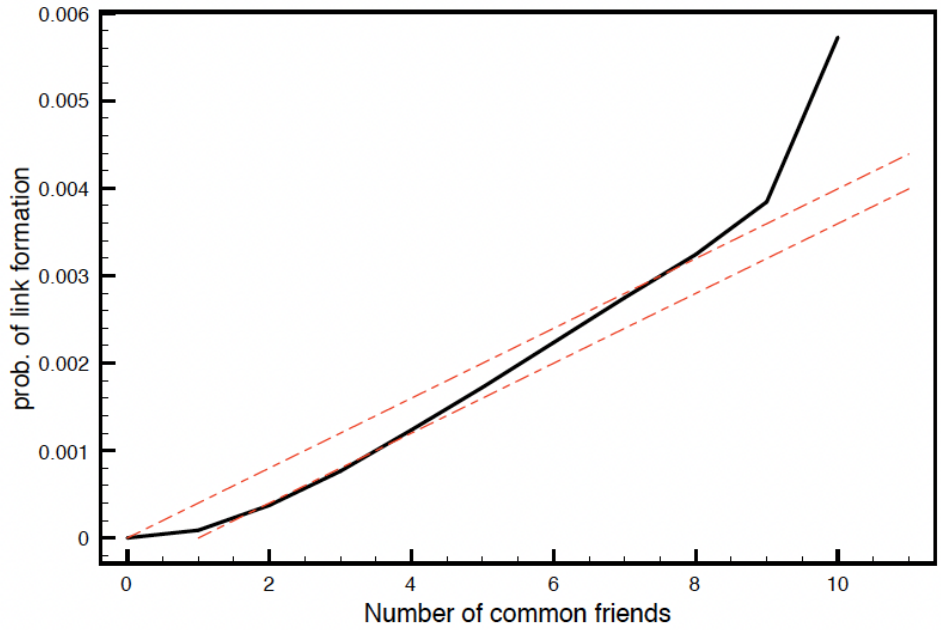
- They then determined an "average" version of $T(k)$ by taking multiple pairs of snapshots
  - They built a curve for $T(k)$ on each pair of snapshots using the procedure described above, and then averaged all the curves they obtained
  - The observations in each snapshot were one day apart, so their computation gives the average probability that two people form a link per day, as a function of the number of common friends they have

Tracking Link Formation: Triadic Closure



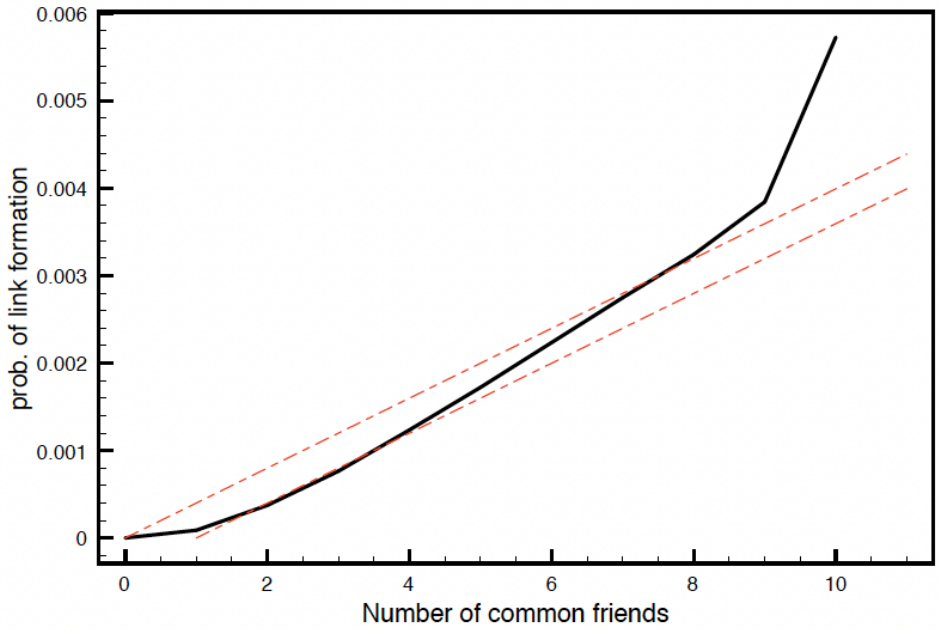
- Clear evidence for triadic closure: $T(0)$ is very close to 0, after which the probability of link formation increases steadily as the number of common friends increases
- Moreover, for much of the plot, this probability increases in a roughly linear fashion as a function of the number of common friends, with an upward bend away from a straight-line shape

Tracking Link Formation: Triadic Closure



- The curve turns upward in a particularly pronounced way from 0 to 1 to 2 friends: having two common friends produces significantly more than twice the effect on link formation compared to having a single common friend

Tracking Link Formation: Triadic Closure



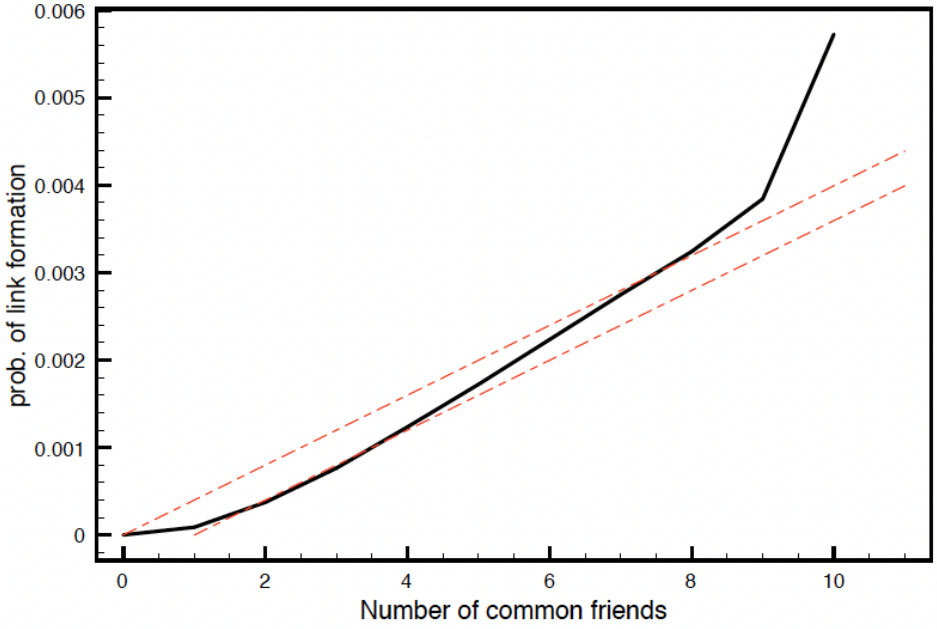
- The upward effect from 8 to 9 to 10 friends is also significant, but it occurs on a much smaller sub-population, since many fewer people in the data have this many friends in common without having already formed a link

Tracking Link Formation: Triadic Closure

- To interpret this plot more deeply, we can compare it to an intentionally simplified baseline model
  - What we might expect the data to look like in the presence of triadic closure
- Suppose that for some small probability p, each common friend that two people have gives them an independent probability p of forming a link each day
- So if two people have k friends in common, the probability they fail to form a link on any given day is $(1-p)^k$
  - Each common friend fails to cause the link to form with probability $(1-p)$
  - These k trials are independent
- Since $(1-p)^k$ is the probability the link fails to form on a given day, the probability that it does form is

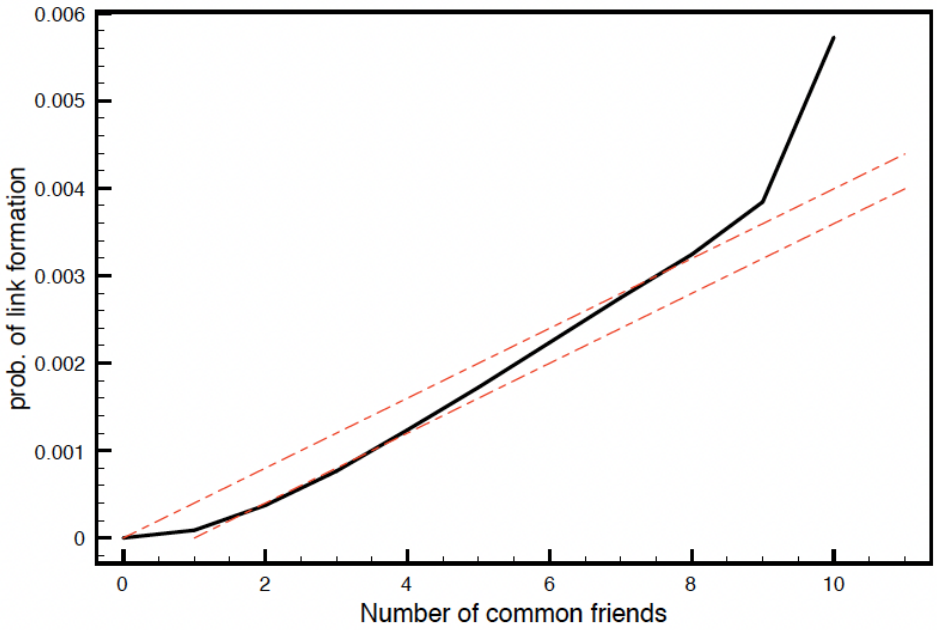
$$T_{baseline}(k) = 1 - (1-p)^k :$$

Tracking Link Formation: Triadic Closure



- We plot this curve in the Figure as the upper dotted line
- Given the small absolute effect of the first common friend in the data, we also show a comparison to the curve $1-(1-p)^{k-1}$, which just shifts the simple baseline curve one unit to the right
- The point is to look at how the real data compares to this baseline

Tracking Link Formation: Triadic Closure



- Both the real curve and the baseline curve are close to linear
- But the real data turns upward
- This indicates that the assumption of independent effects from common friends is too simple to be fully supported by the data

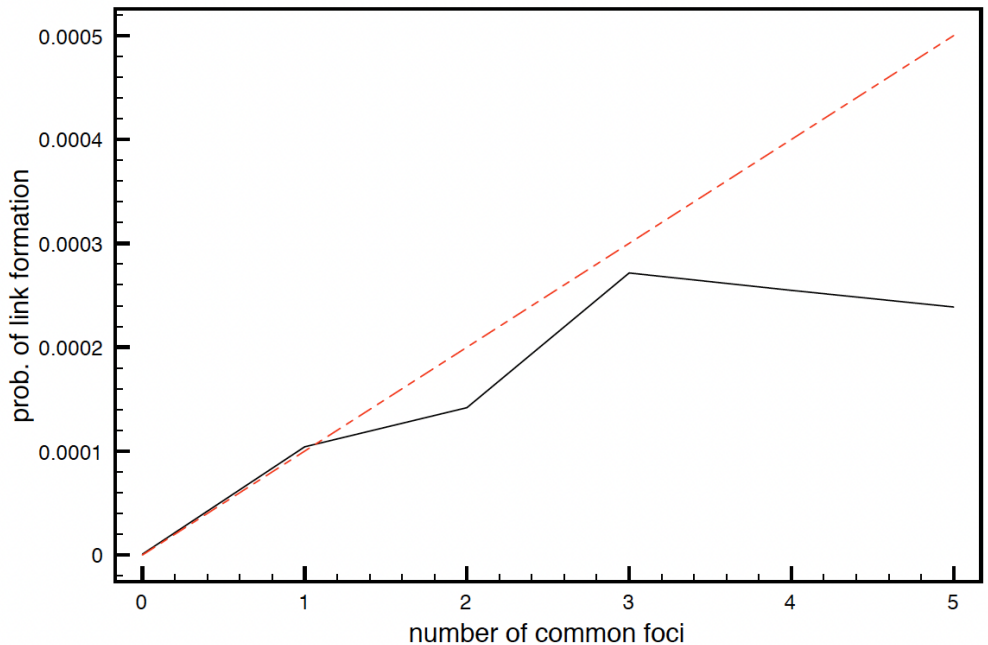
Tracking Link Formation: Focal Closure

What is the probability that two people form a link as a function of the number of foci they are jointly affliated with?

- For focal closure, Kossinets and Watts supplemented their university e-mail dataset with information about the class schedules for each student
- Each class is a focus, and two students shared a focus if they had taken a class together
- How can they compute the probability of focal closure?

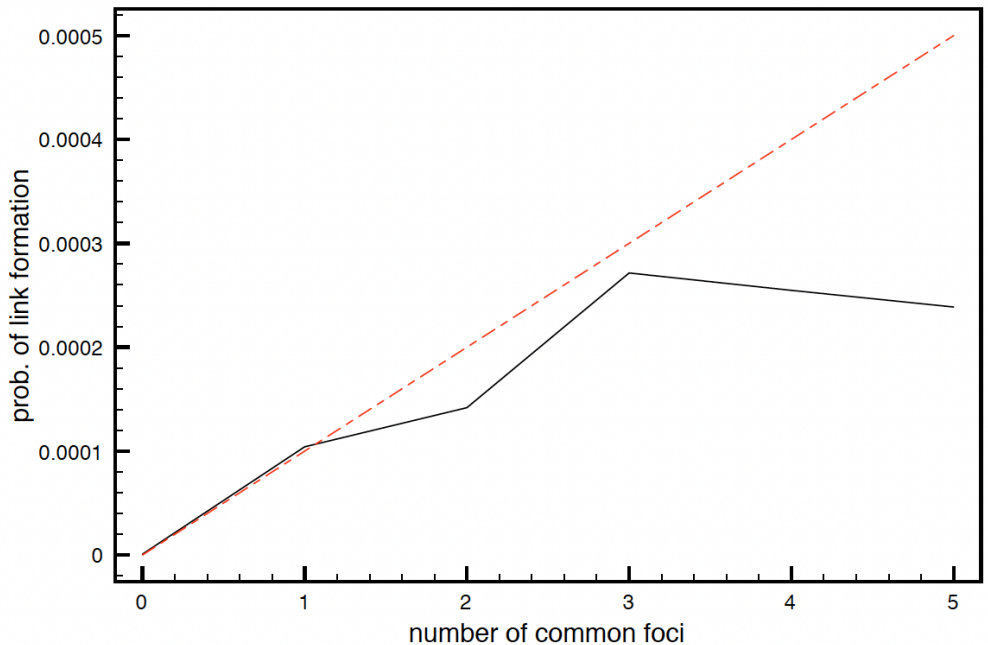
Tracking Link Formation: Focal Closure

They determine the probability of link formation per day as a function of the number of shared foci:

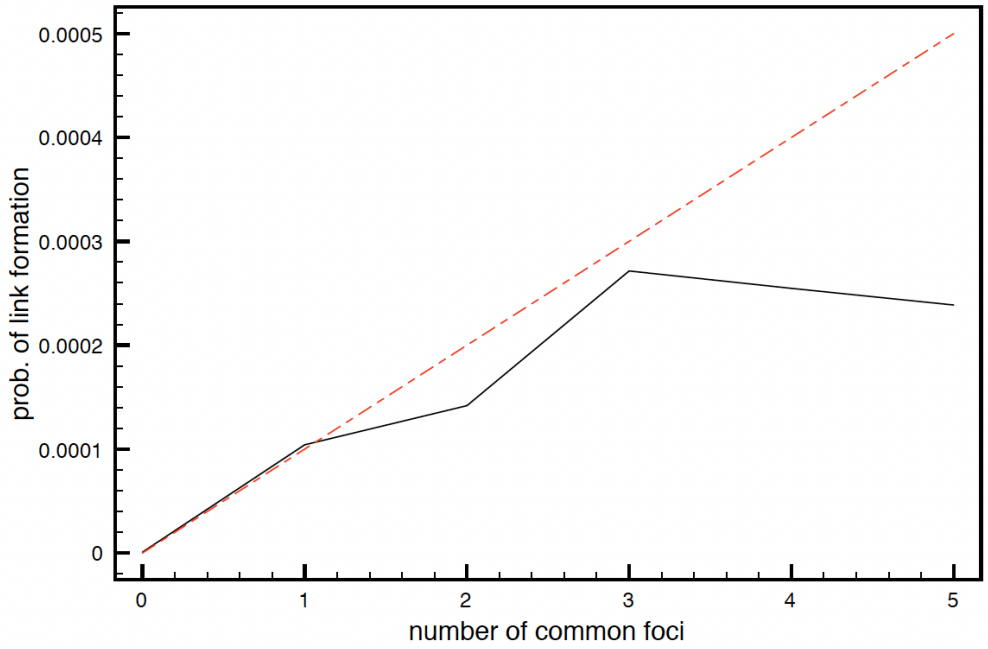


Tracking Link Formation: Focal Closure

A single shared class turns out to have roughly the same absolute effect on link formation as a single shared friend, but..



Tracking Link Formation: Focal Closure



- The curve for focal closure behaves turns downward and appears to approximately level off, rather than turning slightly upward
- Subsequent shared classes after the first produce a "diminishing returns" effect
- Comparing to the same kind of baseline, in which the probability of link formation with k shared classes is $1-(1-p)^k$, we see that the real data turns downward more significantly than this independent model

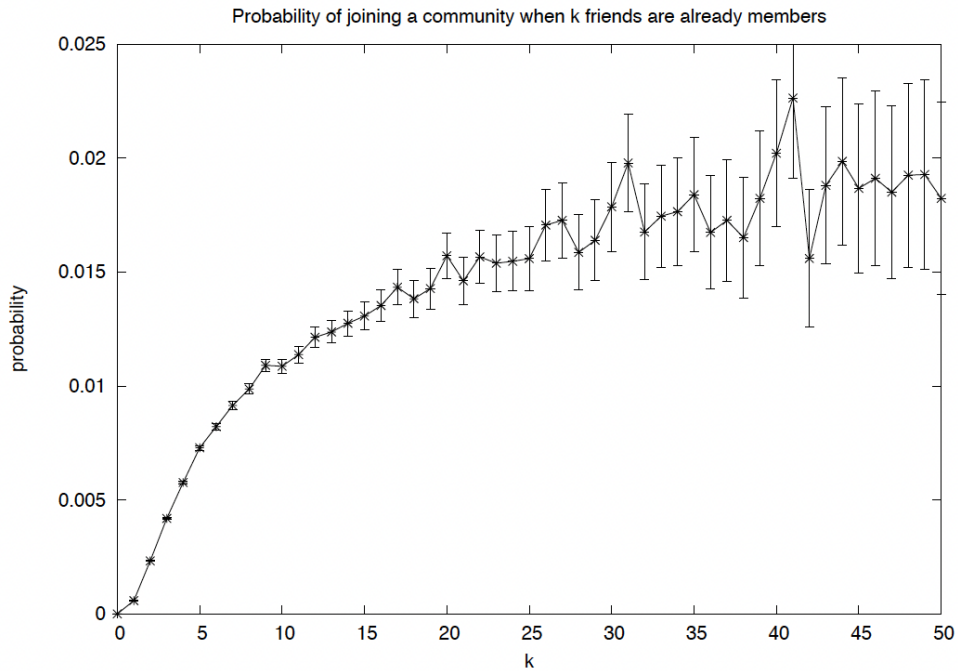
Tracking Link Formation: Membership Closure

What is the probability that a person becomes involved with a particular focus as a function of the number of friends who are already involved in it?

- This has been measured in online domains that possess both person-to-person interactions and person-to-focus affiliations
- Example: Blogging site LiveJournal
  - Friendships = users in their profiles
  - Foci = membership in user-defined communities

Tracking Link Formation: Membership Closure

Probability of joining a community as a function of the number of friends who have already done so:

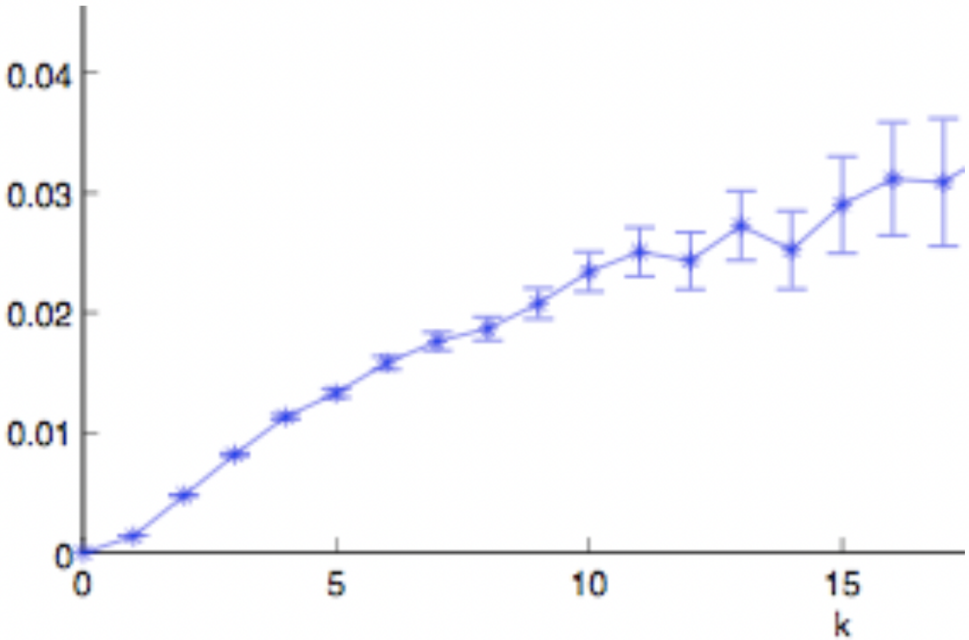


Tracking Link Formation: Membership Closure

- Similar analysis for Wikipedia
- A node for each Wikipedia editor who maintains a user account and user talk page on the system
- There is an edge joining two such editors if they have communicated, with one editor writing on the user talk page of the other
- Each Wikipedia article defines a focus (association if the editor edited the article)

Tracking Link Formation: Membership Closure

Probability a person edits a Wikipedia article as a function of the number of prior editors with whom they have communicated:



Tracking Link Formation: Membership Closure

- The probabilities in both cases increase with the number k of common neighbors
- The marginal effect diminishes as the number of friends increases, but the effect of subsequent friends remains significant
- In both cases, there is an initial increasing effect similar to what we saw with triadic closure
  - The probability of joining a LiveJournal community or editing a Wikipedia article is more than twice as great when you have two connections into the focus rather than one

Interplay between Selection and Social Influence

- We make use of the Wikipedia data: *How do similarities in behavior between two Wikipedia editors relate to their pattern of social interaction over time?*
- We need to define:
  - The social network
  - A measure of behavioral similarity

Interplay between Selection and Social Influence

- The social network will consist of all Wikipedia editors who maintain talk pages, and there is an edge connecting two editors if they have communicated
- An editor's behavior will correspond to the set of articles they have edited
- How can we define the similarity between two editors based on their actions?

Interplay between Selection and Social Influence

- A simple solution is to declare their similarity to be the value of the ratio

$$\frac{\text{number of articles edited by both A and B}}{\text{number of articles edited by at least one of A or B}}$$

- Does this sound familiar?

Interplay between Selection and Social Influence

- The ratio is precisely the neighborhood overlap of two editors in the bipartite affliation network of editors and articles, consisting only of edges from editors to the articles they've edited
- Pairs of Wikipedia editors who have communicated are significantly more similar in their behavior than pairs of Wikipedia editors who have not communicated, so we have a case where homophily is clearly present

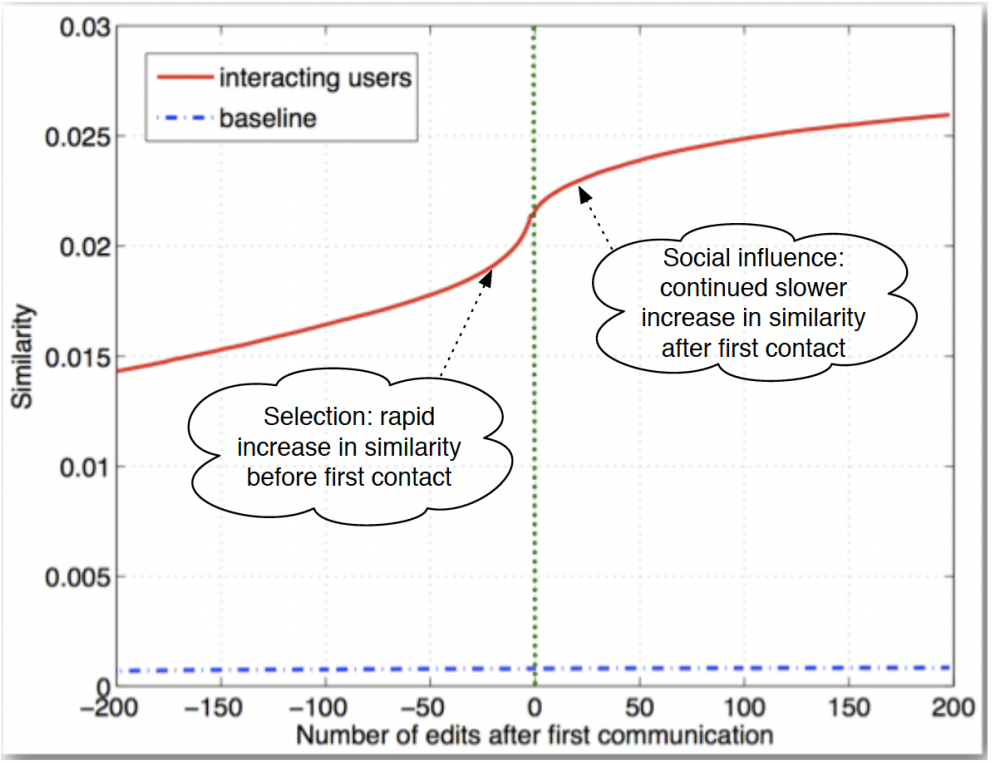
Interplay between Selection and Social Influence

Is the homophily arising because editors are forming connections with those who have edited the same articles they have (selection), or is it because editors are led to the articles of those they talk to (social influence)?

- For each pair of editors A and B who have ever communicated, record their similarity over time (whenever either A or B performs an action on Wikipedia)
- Declare time 0 for the pair A-B to be the point at which they first communicated
- This results in many curves showing similarity as a function of time (one for each pair of editors who ever communicated)

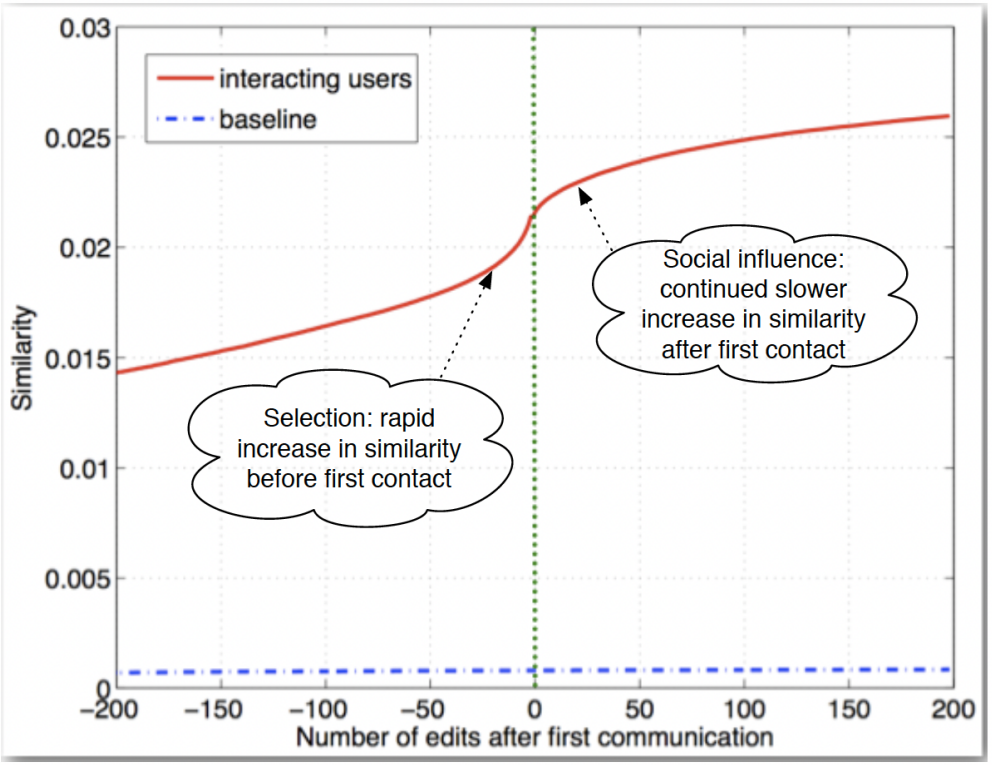
Interplay between Selection and Social Influence

Average level of similarity relative to the time of first interaction, over all pairs of editors who have ever interacted on Wikipedia:



Interplay between Selection and Social Influence

- Similarity is clearly increasing both before and after the moment of first interaction, indicating that both selection and social influence are at work
- However, the curve is not symmetric around time 0
- The levels of similarity depicted in the plot are much higher than for pairs of editors who have not interacted (dashed blue line)



Notes

This presentation is based on Chapter 4 of David Easley and Jon Kleinberg, *Networks, crowds, and markets*, Cambridge University Press, 2012